SAMSUNG

ESSCI
Skilling India in Electronics

PRESENTATION ON

PIZZA SALES REPORT

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CTTC
BHUBANESWAR

Guided by-
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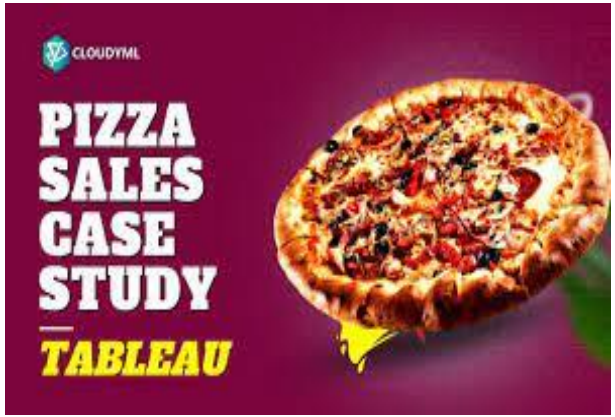
INTRODUCTION

This Pizza Sales Report provides a comprehensive analysis of our pizza sales performance over a specified period. It aims to provide valuable insights into our sales trends, customer preferences, and areas for improvement. By examining various aspects of our pizza sales, we can make informed decisions to enhance our business strategies and boost profitability.

This report provides a comprehensive overview of our pizza sales performance and offers valuable insights to guide our future business decisions. By implementing the recommended strategies, we aim to further enhance our pizza sales and solidify our position in the market.



Project Overview

- Analyze sales trends to identify peak hours, days, and seasons for pizza orders.
- Understand customer preferences by analyzing the most popular pizza varieties and sizes.
- Optimize inventory management based on demand patterns.
- Improve marketing and promotions strategies based on customer behaviour.

Description

The “pizza sales dataset” is a comprehensive collection of data that provides valuable insights into the sales and performance of a fictional pizza restaurant or chain. This dataset is typically used for analytical and statistical purposes, allowing business, researchers, and data enthusiasts to explore various aspects of pizza sales.

Advantages

- **Market Analysis**: The dataset provides valuable insights into the pizza market, helping businesses understand customer preferences, popular pizza types, and trending toppings. This information can inform menu decisions and marketing strategies.

- **Sales Forecasting**: Historical sales data can be used to develop accurate sales forecasts. This is particularly beneficial for optimizing inventory management, staffing levels, and resource allocation.

- **Customer Segmentation**: By analyzing customer data, businesses can segment their customer base based on factors like ordering frequency, preferred pizza types, and demographics. This enables targeted marketing efforts and personalized customer experiences.

- **Menu Optimization**: Analysis of pizza types, sizes, and toppings can inform menu adjustments, allowing businesses to highlight popular items and refine less-performing ones.

Disadvantages

- **Data Quality Issues**: Data quality can be a significant concern. Inaccurate or incomplete entries, missing values, and inconsistencies in data format can affect the reliability of analysis and results.

- **Limited Scope**: The dataset is focused on pizza sales, which may not provide a holistic view of a restaurant's operations. It lacks data on other aspects such as customer service, kitchen operations, and marketing campaigns.

- **Data Privacy and Ethics**: Handling customer data, including customer IDs and delivery addresses, raises privacy and ethical considerations. Ensuring the protection of sensitive information is crucial.

- **Sample Bias**: If the dataset represents only a subset of locations, time periods, or customer segments, it may suffer from sample bias. The findings may not be representative of the entire business or market.

Problem Statement

This problem statement outlines the challenges and objectives faced by the pizza chain, highlighting the need for data-driven solutions and insights derived from the "Pizza Sales Dataset" to drive strategic decision-making and enhance overall business performance.To achieve this goal, the business possesses a comprehensive "Pizza Sales Dataset" that captures various aspects of its operations, including sales transactions, customer feedback, and operational details. However, the business faces several challenges and opportunities that require data-driven solutions.

Following are needed tasks for the development of the algorithm:

1. Download and preprocess the dataset
2. Build and train the model to classify the data
3. Test the model and further improve the model using different techniques.

Data Exploration

Data exploration provides a foundation for more in-depth analysis and helps identify areas of interest and potential questions to address using the "Pizza Sales Dataset." It is an essential step in gaining insights and making data-driven decisions for a pizza restaurant or chain.

Following are the file descriptions and URL’s from which the data can be obtained :

- pizza.zip - zipped folder of all (train/test) images
- sample_submission.csv - a sample submission file in the correct format
- Pizza_sales_list.csv - a list of training images, their subject id, and
- class id

What does it Predict?

- Total revenue, Average order value,Total pizzas sold,Total Orders and Average pizzas per order of pizza sales.
- Hourly trend for pizzas sold.
- Percentage of sales by pizza category.
- Percentage of sales by pizza size.
- Total orders and pizzas sold by pizza category.
- Weekly trend and pizzas sold by pizza category.

Approach

We followed the following four approaches one by one for improving the model results

- Making a Dashboard.

Tableau - Book1 - Tableau licence expires in 5 days

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Show Me

Dashboard | Layout

Selected item
Pizza Tableau 1.jpg

Show title
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Control visibility using value

Position
x 8 y 8

Size
w 1,584 h 884

Border
None

Background
70%

Item hierarchy
- delivery-man.png
- order-delivery.png
- profit-growth.png
- PIZZA SALES REPORT
- KPI BANNER
- Tiled

Total Revenue	Avg Order Value	Total Pizzas Sold	Total Orders	Avg Pizzas Per Order
$666.3K	$38.2	40.4K	17.4K	2

BUSIEST HOURS AND WEEKS

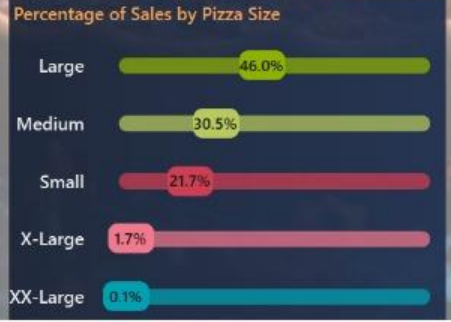
HOURS
Peak orders are between 12:00 PM and 1:00 PM,and in evening from 4:00 PM to 7:00 PM.

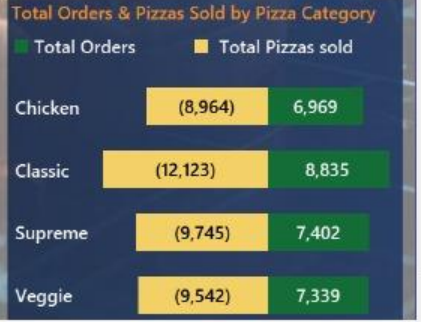
WEEKS
Significant variations in weekly orders with highest peak during the 26th Week

SALES PERFORMANCE

CATEGORY
Classic Category comtributes to maximum Sales,Total Orders and Total Pizzas Sold

SIZE
Large Pizza Size contributes to maximum Total Sales





Data Source | HOME | KPI BANNER | Percentage of Sales by Pizza Ca... | Weeky Trend for Total Orders | Hourly Trend for Total Pizzas Sold | Total Orders & Pizzas Sold by Pi... | Date Range | Percentage (

35°C Partly sunny

Search

ENG IN

16:49 03-09-2023

Tableau - Book1 - Tableau licence expires in 5 days

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

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Dashboard | Layout

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Percentage of Sales by Piz...

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- [ ] Control visibility using value

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Size
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Border
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Background
None

Item hierarchy

- Total Orders & Pizzas Sol...
- Pizza Category
- Hourly Trend for Total Pi...
- Weeky Trend for Total Ord..
- Percentage of Sales by Pi...
- SALES PERFORMANCE

Total Revenue	Avg Order Value	Total Pizzas Sold	Total Orders	Avg Pizzas Per Order
$60.9K	$20.4	3.3K	3.0K	1

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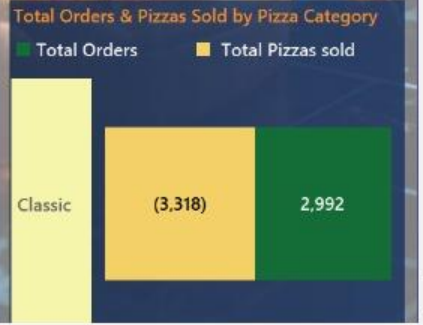

Data Source | HOME | KPI BANNER | Percentage of Sales by Pizza Ca... | Weeky Trend for Total Orders | Hourly Trend for Total Pizzas Sold | Total Orders & Pizzas Sold by Pi... | Date Range | Percentage (

1 of 5 marks 1 row by 1 column SUM of AGG(avg(0)): 0 Highlighting on Pizza Category

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File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Entire View | Show Me

Dashboard | Layout

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Total Orders & Pizzas Sold...

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- [x] Floating
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Border
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Background
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Item hierarchy
- Weekly Trend for Total O...
- Order Date
- Total Orders & Pizzas Sol...
- Measure Names
- Total Orders & Pizzas Sol...
- Pizza Category

Total Revenue	Avg Order Value	Total Pizzas Sold	Total Orders	Avg Pizzas Per Order
$203.0K	$22.3	12.7K	9.1K	1

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Data Source | HOME | KPI BANNER | Percentage of Sales by Pizza Ca... | Weeky Trend for Total Orders | Hourly Trend for Total Pizzas Sold | Total Orders & Pizzas Sold by Pi... | Date Range | Percentage (

1 of 8 marks 4 rows by 1 column SUM of Measure Values: -3,349 Highlighting on Pizza Category

35°C Partly sunny | Search | ENG IN | 16:51 03-09-2023

Tableau - Book1 - Tableau licence expires in 5 days

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Entire View | Show Me

Dashboard | Layout

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KPI BANNER

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- [x] Floating
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Border
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Background
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Item hierarchy
- delivery-man.png
- order-delivery.png
- profit-growth.png
- PIZZA SALES REPORT
- KPI BANNER
- Tiled

Total Revenue	Avg Order Value	Total Pizzas Sold	Total Orders	Avg Pizzas Per Order
$82.8K	$23.5	4.0K	3.5K	1

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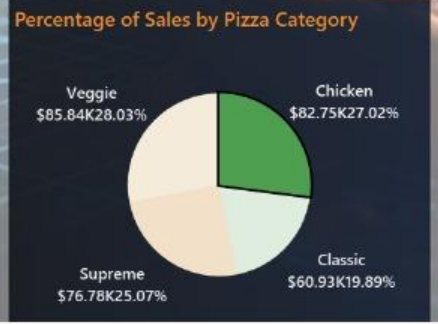

Data Source | HOME | KPI BANNER | Percentage of Sales by Pizza Ca... | Weeky Trend for Total Orders | Hourly Trend for Total Pizzas Sold | Total Orders & Pizzas Sold by Pi... | Date Range | Percentage

1 of 5 marks 1 row by 5 columns SUM of Measure Values: 82,751 Highlighting on Pizza Category

35°C Partly sunny | Search | ENG IN | 16:47 03-09-2023

Conclusion

In conclusion, the "Pizza Sales Dataset" provides a rich source of information for understanding and optimizing the operations of a pizza restaurant or chain. Through data exploration and analysis, several key insights and conclusions can be drawn from this dataset .In essence, the "Pizza Sales Dataset" offers a comprehensive view of a pizza restaurant or chain's operations, customer behavior, and market dynamics. By leveraging the insights derived from this dataset, businesses can make informed strategic decisions, enhance customer experiences, and optimize their overall performance in the highly competitive pizza industry.

Installation

1. Install `Python 3.10` on your Local Machine
2. Create Python Virtual Environment in root folder by opening terminal and executing
   `* pip install`
3. Install Required by python libraries.
   `*pip install pandas and numpy`

4. Install Apache Pyspark.
5. Install MySql and ODBC driver.
6. Install Tableau Desktop (Dashboard)

THANK YOU